National
Qualifications
SPECIMEN ONLY

SQ07/AH/01 Computing Science

Date — Not applicable

Duration — 2 hours

Total marks — 60

Attempt ALL questions

Write your answers clearly in the answer booklet provided. In the answer booklet you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

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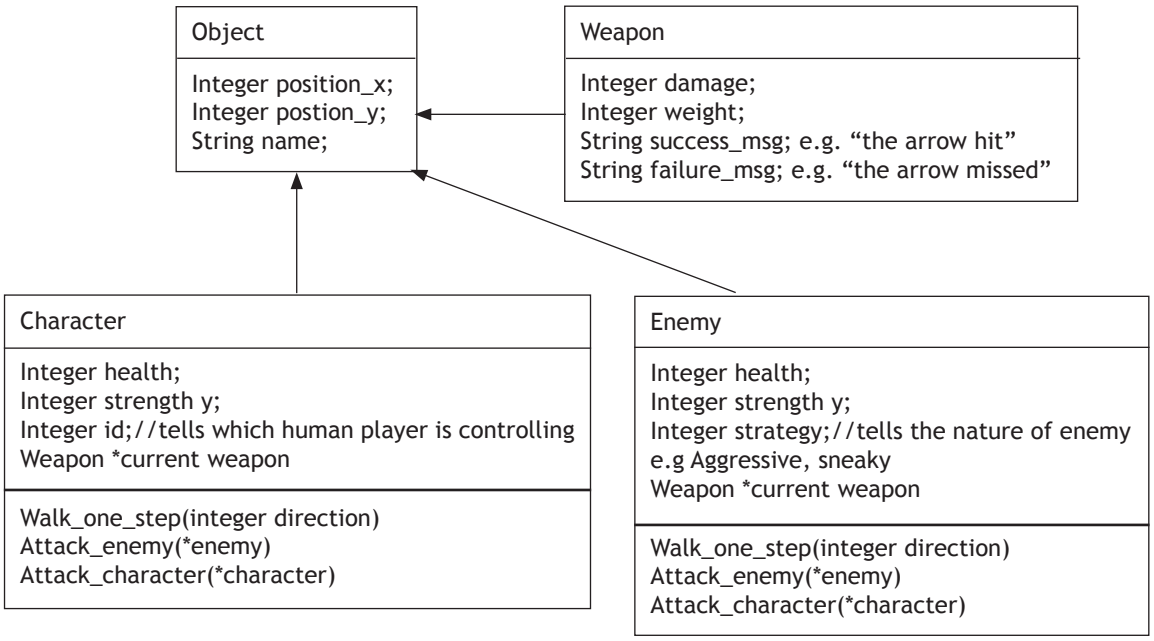
SQ07AH01

MARKS

Attempt ALL questions

1. GamePlay is developing a new computer game. In the game there are many objects including characters, enemies, and weapons. All of the characters in the game can be controlled by the human game players. A class diagram is shown below:



(a) Explain why the `Object` class contains position variables but the other classes do not. **2**

(b) Describe what is meant by encapsulation, with reference to how it has been applied in the `Character` class. **2**

(c) (i) One opinion is that the classes are inefficiently structured. Explain why this could be true. **1**

(ii) Describe how the classes could be restructured to be more efficient. **3**

(d) The program, based on the above classes, contains the following lines of code:

```
1.  SET davina TO <create an object of the Character class>
2.  SET modzilla TO <create an object of the Enemy class>
3.  davina.Attack_enemy(modzilla)
```

The purpose of line 3 is for davina to attack modzilla. Explain, using object-oriented terminology, how this is achieved. **2**

(e) Using an appropriate example, explain how GamePlay could use goal tracking to evaluate the success of the development. **2**

(f) A freelance graphic artist has been contracted by GamePlay to create new characters for the game. Describe how GamePlay would protect its interests when writing a contract for this work. **1**

MARKS

1. (continued)

(g) Players of the new game will be able to take part in online tournaments hosted on the GamePlay website.

Gameplay gathers information automatically about its users during online game play. Explain how this tracking information could be used by GamePlay to influence the development of games in the future. **2**

MARKS

2. The Biskit Box is developing a website to display details of the biscuit boxes that they sell. Customers will register on the website before placing orders.

(a) Part of the HTML used for the search box is shown below:

```
1.  <form name="searchform" action="……" method="……" >
2.  <table>
3.    <tr>
4.      <td><img src= "search_icon.gif" /></td>
5.      <td><input type="……" name="criteria"
6.      value="……"></td>
7.      <td><input type="submit" name="……" value="GO"></td>
8.    </tr>
9.  <table>
    </form>
```

A customer using the search facility sees the following string in the address bar of her browser:

http://thebiskitbox.co.uk/search_script.php?criteria=White+Chocolate&searchbox=GO

(i) What method of processing would be used in line 1? Explain your answer. **2**

(ii) State the missing contents of the `type and value` attributes needed to complete line 5. **2**

(iii) State the missing contents of the `name` attribute needed to complete line 6. **1**

MARKS

2. (continued)

(b) Details of the Customers, BiskitBoxes and Orders will be stored in the following four relations:

```
Customer(CustomerRef, CustomerName, Address, Postcode, Email)
BiskitBox(BoxCode, Description, Price, StockLevel)
Order(OrderID, CustomerRef, OrderDate)
OrderLine(OrderID, BoxCode, QuantityOrdered)
```

Note: In this representation, the primary key of each relation has been underlined.

When an order is placed, a customer-order process is initiated. This uses server-side script to display a summary of all items in the order. The summary should display only the BoxCode, Description, Price and QuantityOrdered, as shown below:

BoxCode	Description	Price	QuantityOrdered
TCZ2	Double Choc	£6.95	1
TCZ3	Oat Biscuit	£6.25	1
TCZ5	Iced Biscuit	£5.49	2

(i) Write a SQL query that will extract the details needed to display the summary for OrderID 759. **4**

During the development of the website, testing of the customer-order process is carried out. This testing highlights the need to display the summary details above in decreasing order of Price.

(ii) Integrative testing is needed to ensure that the separate components of a system work correctly. Describe how testing of this customer-order process is integrative. **2**

(iii) Write the SQL statement that must be **added** to your query in part (b) (i), to display the summary details in decreasing order of Price. **1**

(c) After the website has been in use for several months, the management of The Biskit Box decides that the system should record the CardPaymentNumber used by the Customer to pay for an Order.

State the type of maintenance made necessary by this decision. **1**

(d) The Biskit Box has created a new biscuit range. Explain **two** ways analytics could be used to inform decisions about an online marketing campaign for this new range. **2**

MARKS

3. A computerised version of a card game, based on various animals native to Scotland, is being developed for a website.

During game play, players can take a card from or place a card on a pile of cards. A stack data structure will represent this pile of cards.

(a) The stack is held in a 1-D array and the last item placed in the stack was the Golden Eagle. The 1-D array in which the stack is held is shown below:

Index	Character
0	Ptarmigan
1	Otter
2	Golden Eagle
3	
4	

(i) An item is added to the stack by "pushing" and removed by "popping".

Draw the final state of the stack after the following five operations:

1. Pop
2. Push Loch Ness Monster
3. Pop
4. Pop
5. Push Grouse **2**

(ii) Apart from the 1-D array, describe another item of data required to implement a stack. **1**

(iii) When a stack is implemented using a 1-D array, adding a valid item can cause an execution error.

Explain why an execution error can occur in this situation. **2**

(b) A linked list could have been used rather than a stack to represent the pile of cards.

(i) The animals Ptarmigan, Otter and Golden Eagle are entered, in the order given, into a linked list. Draw this linked list. **2**

(ii) Explain why the execution error in part (a) (iii) would not occur if a linked list is used rather than a stack. **2**

3. (continued) **MARKS**

(c) The game includes a high score table with the names and scores of the top five players stored in two 1-D arrays.

High Score Table

Name	Score
Meena	58
Joe	50
Patrick	27
Marta	25
Andrew	23

After a number of games, the 1-D array that holds the scores is shown below:

Index	0	1	2	3	4
Scores	23	25	27	50	58

Senga plays the game and scores 48. Her score of 48 and her name now replace the lowest score and name that were in position 0. The 1-D array for the scores is now:

Index	0	1	2	3	4
Scores	48	25	27	50	58

A bubble sort is used to sort the scores. After the first pass, the list will be sorted:

Index	0	1	2	3	4
Scores	25	27	48	50	58

(i) State the **two** exchanges that took place in this pass. **2**

(ii) Explain why the bubble sort will make another pass through the list, even though it is sorted. **2**

(iii) The updated high score table is displayed as shown.

High Score Table

Name	Score
Meena	58
Joe	50
Patrick	48
Marta	27
Senga	25

Identify the error in the table and explain one possible cause of this error. **2**

MARKS

4. A file holds the timetable for a bus route.

The first line of the file contains names of all the bus stops on the route, in order, separated by commas.

Each subsequent line of the file contains stopping times for a particular service. Note that:

- these times are comma-separated in hh:mm, 24-hour clock format
- the first time corresponds to the first stop on the route, the second time to the second stop and so on
- if a service does not stop at a particular stop, then "--:--" is used as a place holder for the stopping time.

Part of the file is shown below:

```
Park Place,Union Street,Clyde Bowling Green,West College
10:00, 10:05, 10:16, 10:21
10:15, --:--, 10:31, 10:36
……————, ……————, ……————, ……———— and so on
```

This shows details of two services on a bus route that has four stops. Notice that the second service doesn't stop at Union Street.

A program is required that enables the user to specify a particular bus stop and a time. The program should find the time at which the first service stops after the time entered:

For example, if the specified stop was "Park Place" and the time entered was 10:12, then the expected output for the sample data above would be 10:15.

(a) Define one or more data structures to hold the bus timetable information, explaining how the various parts of the timetable would be stored in the data structure. **3**

The simple textual interface for testing the system is shown below:

```
                    Bus Timetable System

Bus Stops on Route:
1.Park Place
2.Union Street
3.Clyde Bowling Green
4.West College

Please type the number of the stop you want to use (-1 to finish):      3
Please enter the departure time between 06:30 and 23:30:            10:20
The first service from Clyde Bowling Green departs at 10:31:
Please type the number of the stop you want to use (-1 to finish):     -1
```

MARKS

4. (continued)

(b) The top level design for the program is shown below:

1. Create data structure(s) to store timetable data
2. Read in timetable data from file and store in data structure
3. Process user requests

(i) Using pseudocode, or any other design notation, refine step 2 of the top level design. Make use of the data structure(s) you defined in part (a).

You may assume that there are no more than 20 stops and 15 services in this bus route. **4**

(ii) The refinement of step 3 of the top level design begins:

```
3.1  Display welcome message
3.2  Display list of bus stops
3.3  Ask for bus stop number
3.4  While bus stop number ≠ 1
3.5     Ask for and validate departure time
3.6     Find suitable bus
3.7     Display arrival time of next bus
3.8     Ask for bus stop number
3.9  End while
```

Using a programming language with which you are familiar, code a subprogram that could be used to implement step 3.6 of this design.

Your solution should make use of the data structure(s) you defined in part (a) and should indicate any parameters that would be required.

As before, you may assume there are no more than 20 stops and 15 services.

You may also assume that buses on this service run between the hours of 06:30 and 23:30 inclusive. **8**

[END OF SPECIMEN QUESTION PAPER]

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SQ07/AH/01 **Computing Science**

Marking Instructions

These Marking Instructions have been provided to show how SQA would mark this Specimen Question Paper.

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General Marking Principles for Advanced Higher Computing Science

This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this Paper. These principles must be read in conjunction with the Detailed Marking Instructions, which identify the key features required in candidate responses.

(a) Marks for each candidate response must always be assigned in line with these General Marking Principles and the Detailed Marking Instructions for this assessment.

(b) Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.

(c) Marks should be awarded regardless of spelling, as long as the meaning is unambiguous.

(d) Candidates may answer programming questions in any appropriate programming language or pseudocode. Marks should be awarded, regardless of minor syntax errors, as long as the intention of the coding is clear.

(e) Where a question asks the candidate to **describe**, the candidate must provide a statement or structure of characteristics and/or features. This should be more than an outline or a list. It may refer to, for instance, a concept, process, experiment, situation or facts in the context of, and appropriate to, the question. The candidates will normally be required to make the same number of factual/appropriate points as there are marks available for the question.

(f) Where a question asks the candidate to **explain**, marks should only be awarded where the candidate goes beyond a description, for example, by giving a reason, or relating cause to effect, or providing a relationship between two aspects. These will be related to the context of the question or a specific area within a question.

(g) Credit should be given where a labelled diagram conveys clearly and correctly the response required by the question.

Detailed Marking Instructions for each question

Question			Expected response	Max mark	Additional guidance
1	a		Since these are subclasses of the object class **(1 mark),** then these properties are inherited. **(1 mark)**	2	
1	b		The object properties would be hidden and inaccessible outside the Character class or subclass. (**1 mark**) Code within the methods of the class, such as Walk_one_step, can manipulate the properties of an instance of that class, ie the position coordinates of that instance. **(1 mark)**	2	If no reference to character class or methods is given in both parts, then only **1 mark** should be given.
1	c	(i)	Repetition of many lines of code, ie methods and properties in both the character and enemy subclasses.	1	
1	c	(ii)	Create a subclass, eg agents, and place all common properties and methods in that subclass. **(1 mark)** Create two new subclasses of agents, ie character and enemy **(1 mark),** with only their unique property, ie strategy for enemy and id for character. **(1 mark)**	3	Full marks can only be given if reference is made specifically to the unique properties of Object, Character and Enemy.
1	d		The Attack_enemy method of the davina object is called, with modzilla as the actual parameter.	2	Any **two** of the following: method call (**1 mark**) of davina object (**1 mark**) modzilla is actual parameter (**1 mark**)
1	e		GamePlay could set a goal such as "achieve sales of 1000 games" per month **(1 mark)** Once the game had been released, analysis of monthly sales figures would indicate whether or not the goals had been achieved. The achievement of (or failure to reach) this goal would determine the success of the development. **(1 mark)**	2	Answer needs to refer to an example of an appropriate goal.
1	f		GamePlay would need to ensure it owns/has defined use of any intellectual property rights in relation to the work contracted to this freelance artist.	1	
1	g		Cookies could be used to gather data about players' profiles, games they play, and the frequency of their visits to the website. **(1 mark)** Analysis of this data will indicate what types of games are popular with certain age groups and allow the business to focus developments in areas that are likely to generate the most income. **(1 mark)**	2	

Question			Expected response	Max mark	Additional guidance
2	a	i	GET **(1 mark)** The search criteria appear in the browser address bar. **(1 mark)**	2	
2	a	ii	`type="text"` **(1 mark)** `value="Search or enter catalogue number"` **(1 mark)**	2	
2	a	iii	`name="searchbox"`	1	
2	b	i	`SELECT BoxCode, Description, Price, QuantityOrdered` **(1 mark)** `FROM BiskitBox, OrderLine` **(1 mark)** `WHERE OrderID = 759` **(1 mark)** `AND BiskitBox.BoxCode = OrderLine.BoxCode;` **(1 mark)**	4	
2	b	ii	The customer order process would require a server-side script to activate the SQL instruction which would be used to extract the required details from the database **(1 mark)**. Since two separate technologies are being used, integrative testing would be needed to make sure that they connect and communicate correctly. **(1 mark)**	2	
2	b	iii	`ORDER BY Price DESC`	1	
2	c		Perfective maintenance is needed to improve an existing system.	1	
2	d		For example: • evaluating visitor stats from the company's own website or its social media sites to: – enhance search engine rankings – help decide the most effective product placement points and links • using social media analytics to: – determine the most productive links to groups – evaluate the success of advertising placement – assess success of previous competitions or launches	2	Response should indicate **two** different forms of analytics and how they are used.
3	a	i	Ptarmigan at position 0 and Grouse at position 1.	2	
3	a	ii	Stack pointer/integer variable, used to store the **index/position** of current top value of the stack/next available space.	1	

Question			Expected response	Max mark	Additional guidance
3	a	iii	Stack is full or stack overflow. **(1 mark)** **or** Attempting to push an item to a full stack. Attempting to access an index value past the end of the array. **(1 mark)** or An array is of a fixed size — attempting to exceed this index will give an out of range error.	2	
3	b	i	Ptarmigan → Otter → Golden Eagle → Null Correct ordering of list items **(1 mark)** Correct use of pointers and null (tail) **(1 mark)**	2	
3	b	ii	Linked list is a dynamic structure; number of items in list is not fixed. **(1 mark)** Items are not stored contiguously in memory, making it easier to add new items. **(1 mark)**	2	
3	c	i	48 and 25 (values at index 0 and 1). **(1 mark)** 48 and 27 (values at index 1 and 2). **(1 mark)**	2	
3	c	ii	A swap has taken place (and so the Boolean variable will be true). **(1 mark)** The terminating condition of no swaps will not have been met until the end of the second pass. **(1 mark)**	2	
3	c	iii	Senga's name is in the wrong position **(1 mark),** because the algorithm does not swap the names when the scores are swapped. **(1 mark)**	2	
4	a		Any **one** of the following: • A list of lists: the first list in the list of lists would hold the bus stop names as strings, with subsequent lists holding the times of a particular service, also as strings. • A 2D array of strings: the bus stop names stored in row 0 and the times for each service in each subsequent row. • Parallel 1D arrays of strings: the bus stop names stored in element 0 of each array and the times stored in subsequent elements. • Other valid answers are possible.	3	**1 mark** for an appropriate structure. **1 mark** for an explanation of how data structure would store bus stop names. **1 mark** for an explanation of how data structure would be used to store service details.

Question			Expected response	Max mark	Additional guidance
4	b	i	2.1 open file for read 2.2 read data in first line of file 2.3 count number of bus stops (or columns of data) 2.4 for each comma separated entry in line 2.5 store entry in serviceTimes[0,column] 2.6 next column 2.7 set numberOfServices to 0 2.8 for each subsequent line until end of file 2.9 add 1 to number of services 2.10 read data item and store in serviceTimes[numberOfServices,column] 2.11 next column 2.12 close file	4	Candidate answer must refer to data structure(s) defined in part (a) (the sample exemplifies use of a 2D array). **1 mark** for open/close file. **1 mark** for read/store bus stop names. **1 mark** for read/store service times **1 mark** for appropriate reference to data structure(s) defined in part (a).

4	b	ii	function findNextBus sTimes, nServices, bsNumber, dTime
			// 2D array of service times, number of services, bus stop number required and required departure time passed as IN parameters **(1 mark)**
			
			local row, found, position
			// local, variables to indicate row of array being examined, Boolean variable and position of first bus in the array
			
			put false into found
			put 0 into row
			
			repeat until found = true **(1 mark)**
			  if sTimes[row,(bsNumber-1)]="--:--" then **(1 mark)**
			  add 1 to row **(1 mark)**
			  else if sTimes[row,(bsNumber-1)]>=dTime then **(1 mark)**
			  put true into found **(1 mark)**
			  put row into position **(1 mark)**
			  add 1 to row
			  end if
			end repeat
			return sTimes[position,(bsNumber-1)] **(1 mark)**
			// time returned to main program (OUT parameter if procedure is used)
			end function
				8	Candidate answer can be written in any programming language (sample answer is written in LiveCode).
					
					1 mark each should be awarded for correctly indicating:
					• IN parameters required
					• OUT parameter or returned value (function)
					• use of conditional loop
					• check for "--:--"
					• check for required departure time
					• increment array index
					• assign true when first departure time found
					• assign array index to position when first departure time found

[END OF SPECIMEN MARKING INSTRUCTIONS]

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Change since last published:

Change of format from a question paper to question paper with separate answer booklet.